JAVA PROGRAMMING ASSIGNMENT - 1

Subject: Object Oriented Programming

Total Marks: 100

Student Name: ____

Register Number: ______________________________________

Department: Computer Science and Engineering

Semester / Term: Semester 5 - 2026

1. Explain the four fundamental principles of Object-Oriented Programming.

Encapsulation combines data and methods inside a class and controls access to data. Inheritance allows a child class to reuse properties and methods of a parent class. Polymorphism allows one interface to have multiple forms; method overloading and overriding are common examples. Abstraction hides implementation details and exposes only essential features using abstract classes and interfaces.

2. What is a class and what is an object in Java?

A class is a blueprint that defines data members and methods. An object is an instance of a class created at runtime.

```
class Student {
    String name;
    int age;

    void display() {
        System.out.println(name);
        System.out.println(age);
    }

    public static void main(String[] args) {
        Student s = new Student();
        s.name = "Narmatha";
        s.age = 20;
        s.display();
    }
}
```

3. What is inheritance? Explain the types supported in Java.

Inheritance allows one class to acquire properties and methods of another class using extends. Java supports single, multilevel, and hierarchical inheritance through classes. Multiple inheritance of type can be achieved using interfaces.

4. Differentiate between method overloading and method overriding.

Overloading means methods have the same name but different parameter lists in the same class and is compile-time polymorphism. Overriding means a child class provides a new implementation of a parent method and is runtime polymorphism.

5. Explain constructors in Java.

A constructor initializes an object. It has the same name as the class and no return type. A default constructor has no parameters, while a parameterized constructor accepts values to initialize object fields.

6. Difference between abstract class and interface.

An abstract class can contain abstract and concrete methods, constructors, and instance variables. An interface defines a contract for implementing classes. A class can extend one abstract class but can implement multiple interfaces.

7. Write a Java program to create a Student class.

Example program:

```
class Student {
    String name;
    int age;

    Student(String name, int age) {
        this.name = name;
        this.age = age;
    }

    void display() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }

    public static void main(String[] args) {
        Student s = new Student("Narmatha", 20);
        s.display();
    }
}
```

8. Explain exception handling in Java.

Exception handling manages runtime errors. try contains risky code, catch handles exceptions, finally executes cleanup code, throw explicitly raises an exception, and throws declares exceptions that a method may pass to its caller.

9. Write a Java program to demonstrate inheritance and method overriding.

Example program:

```
class Animal {
    void sound() {
        System.out.println("Animal makes a sound");
    }
}

class Dog extends Animal {
    @Override
    void sound() {
        System.out.println("Dog barks");
    }

    public static void main(String[] args) {
        Animal a = new Dog();
        a.sound();
    }
}
```

10. Explain the advantages of OOP and how Java supports OOP.

OOP provides reusability, modularity, maintainability, security, flexibility, and reduced complexity. Java supports OOP through classes and objects, inheritance, encapsulation with access modifiers, polymorphism through overloading and overriding, and abstraction using abstract classes and interfaces.

Declaration: I confirm that the answers submitted in this assignment are my own work and follow the assignment guidelines.

Student Signature: ____________________________ Date: ___________________